

Danilo Giacometti

Home

Research

Publications

Teaching

News

Gallery

News

News

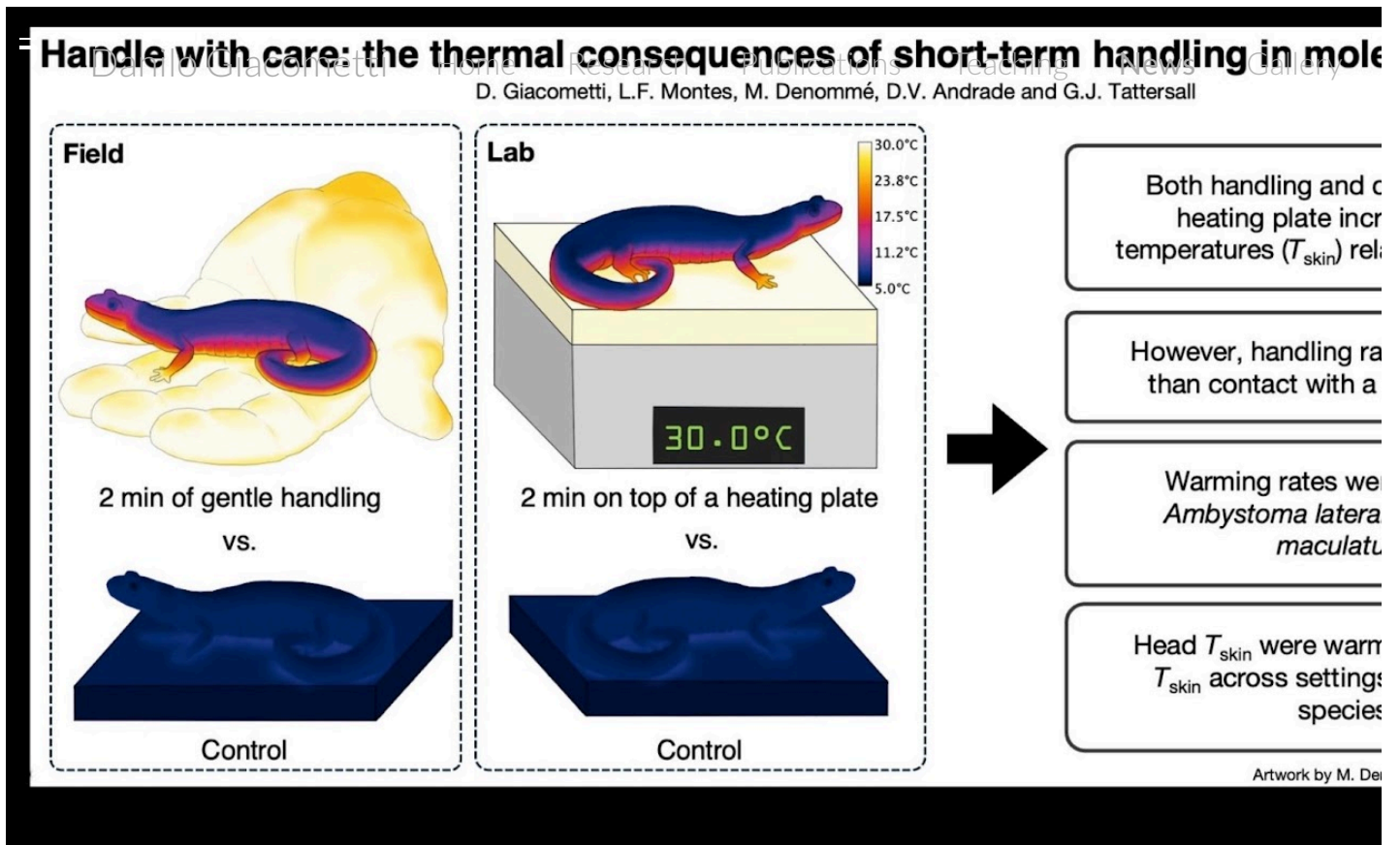




The Company of **Biologists**

February, 2026

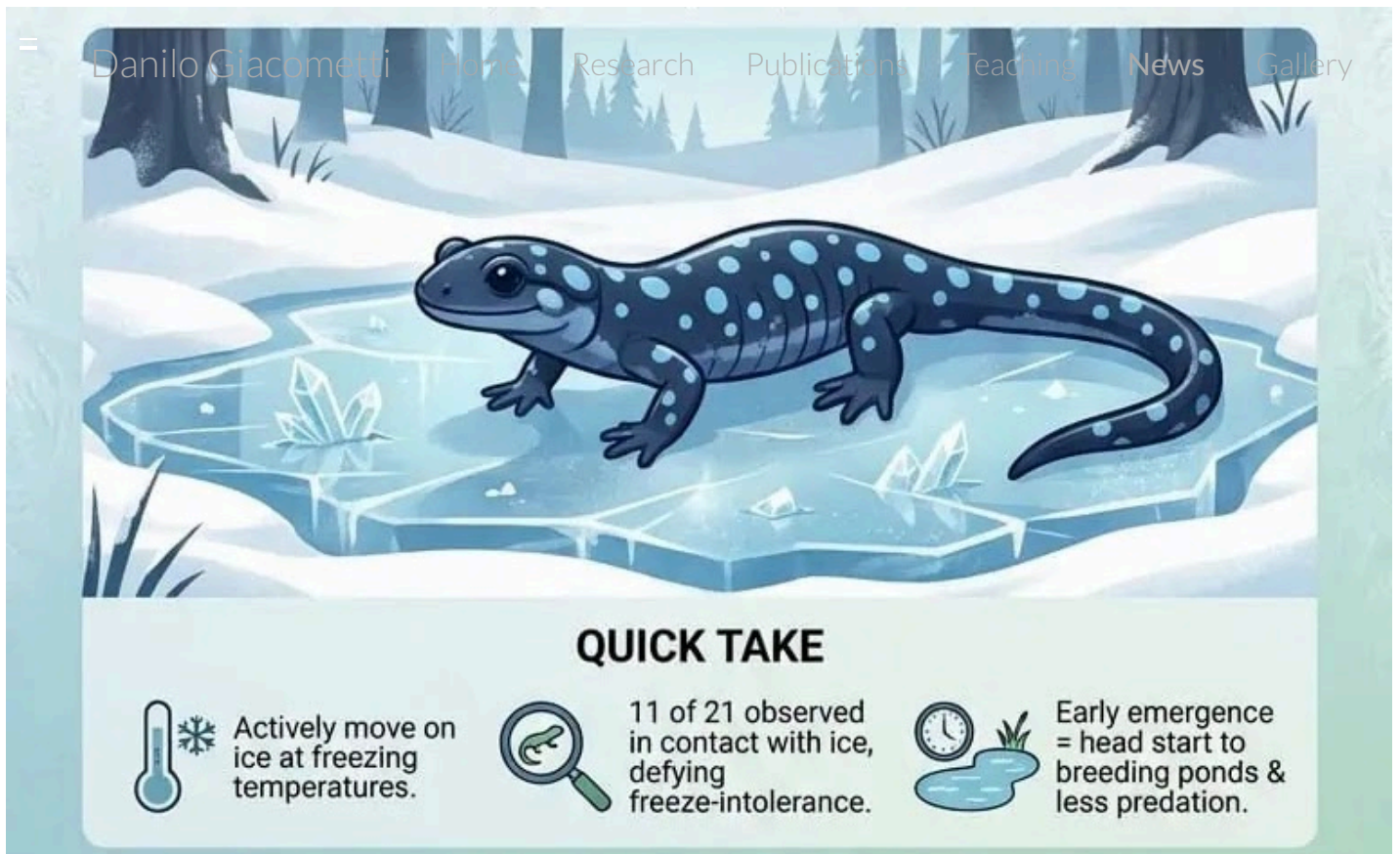
I'm happy to share that our paper made the shortlist for the 2025 Journal of Experimental Biology Outstanding Paper Prize Award. It's an honour to receive such recognition.



January, 2026

New paper out!

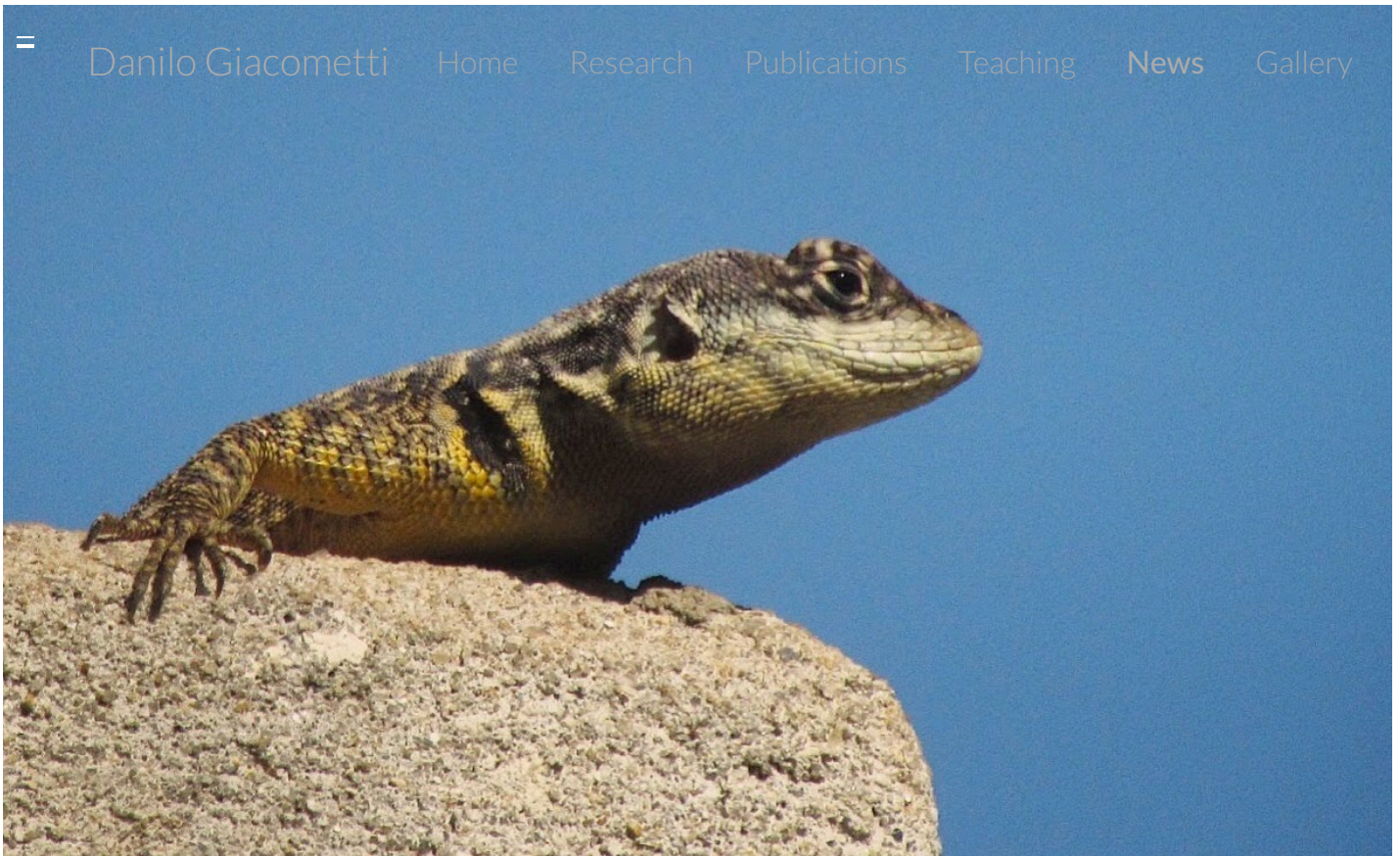
We show that even short-term, gentle handling changes skin temperature and activity in mole salamanders, with implications for thermal physiology research and animal welfare.



January, 2026

Media coverage

Recently, our paper about supercooling in blue-spotted salamanders has caught the eye of NPR and A-Z Animals. Check out the scientific communication articles [here](#), [here](#), and [here](#).



September, 2025

New paper out!

In *Tropidurus catalanensis*, sprint speed isn't about size, testosterone or muscle enzyme activity. Instead, fast lizards show greater cardiac oxidative capacity, which is vital for post-exercise recovery





September, 2025

New paper out!

We demonstrate that sexually selected traits like horns and claws aren't just about mating success: these structures can also shape thermal biology. Photo: A. V. Palaoro





September, 2025

New paper out!

In this paper, we present thermal imaging evidence that blue-spotted salamanders partake in breeding migration at sub-zero body temperatures that are way below their known supercooling point.



[Danilo Giacometti](#)[Home](#)[Research](#)[Publications](#)[Teaching](#)[News](#)[Gallery](#)

**FUNDAÇÃO DE AMPARO À PESQUISA
DO ESTADO DE SÃO PAULO**

July, 2025

I am happy to share that as of August 2025 I will be starting a post-doc position in Dr. Carlos Navas' lab at Universidade de São Paulo funded by the São Paulo Research Foundation.





June, 2025

I just passed my PhD defence! Thank you to my supervisor, Dr. Glenn Tattersall, my examiners, Dr. Donald Miles (Ohio University) and Dr. Toby Mündel, my committee members, Dr. Kiyoko Gotanda and Dr. Miriam Richards, and the chair of the defence, Dr. Diane Mack. Thanks as well to everyone who attended—whether in person or remotely.





May, 2025

New paper out!

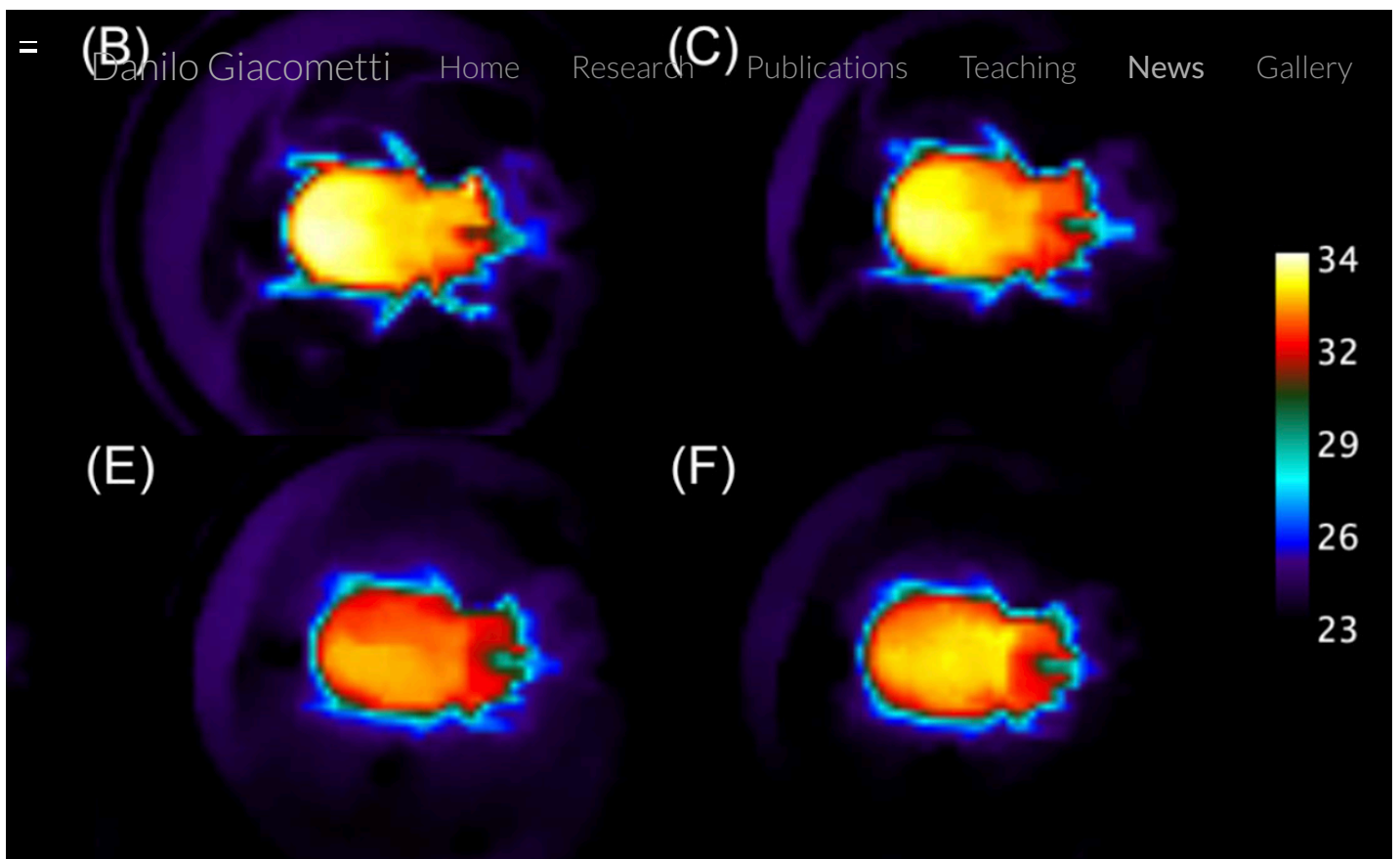
We show that salamanders behaviourally regulate a constant vapour pressure deficit regardless of temperature, which supports the presence of a humidistat that helps regulate evaporative water loss. Make sure to also check out the Inside JEB article about our study here.



April, 2025

New paper out!

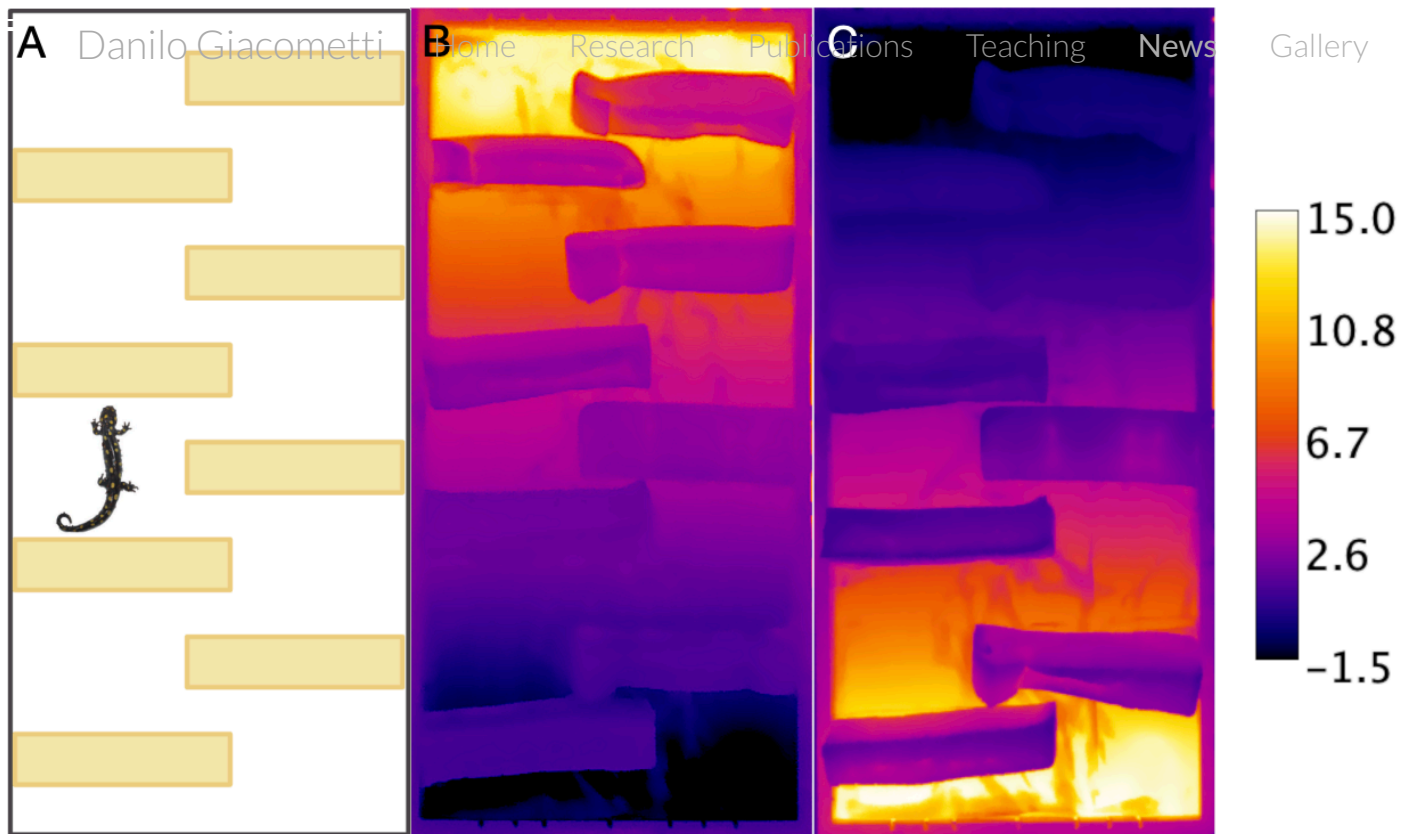
We demonstrate that fossorial (i.e., underground-dwelling) ectotherms, which are thought to experience attenuated climatic variability underground, also display seasonal adjustments in key physiological functions.



March, 2025

New paper out!

With rhinoceros beetles rescued from an international smuggler in São Paulo, Brasil, we used infrared thermography to demonstrate that regional heterothermy can be sustained even in the absence of heat stress or enforced exercise.



October, 2024

New paper out!

In this study, we evaluated how both innate (migration restlessness) and external (temperature and gravity) factors can explain the behaviour of the spotted salamanders emerging from underground burrows at the end of winter. You can also read an ECR Spotlight about me [here](#).





September, 2024

New paper out!

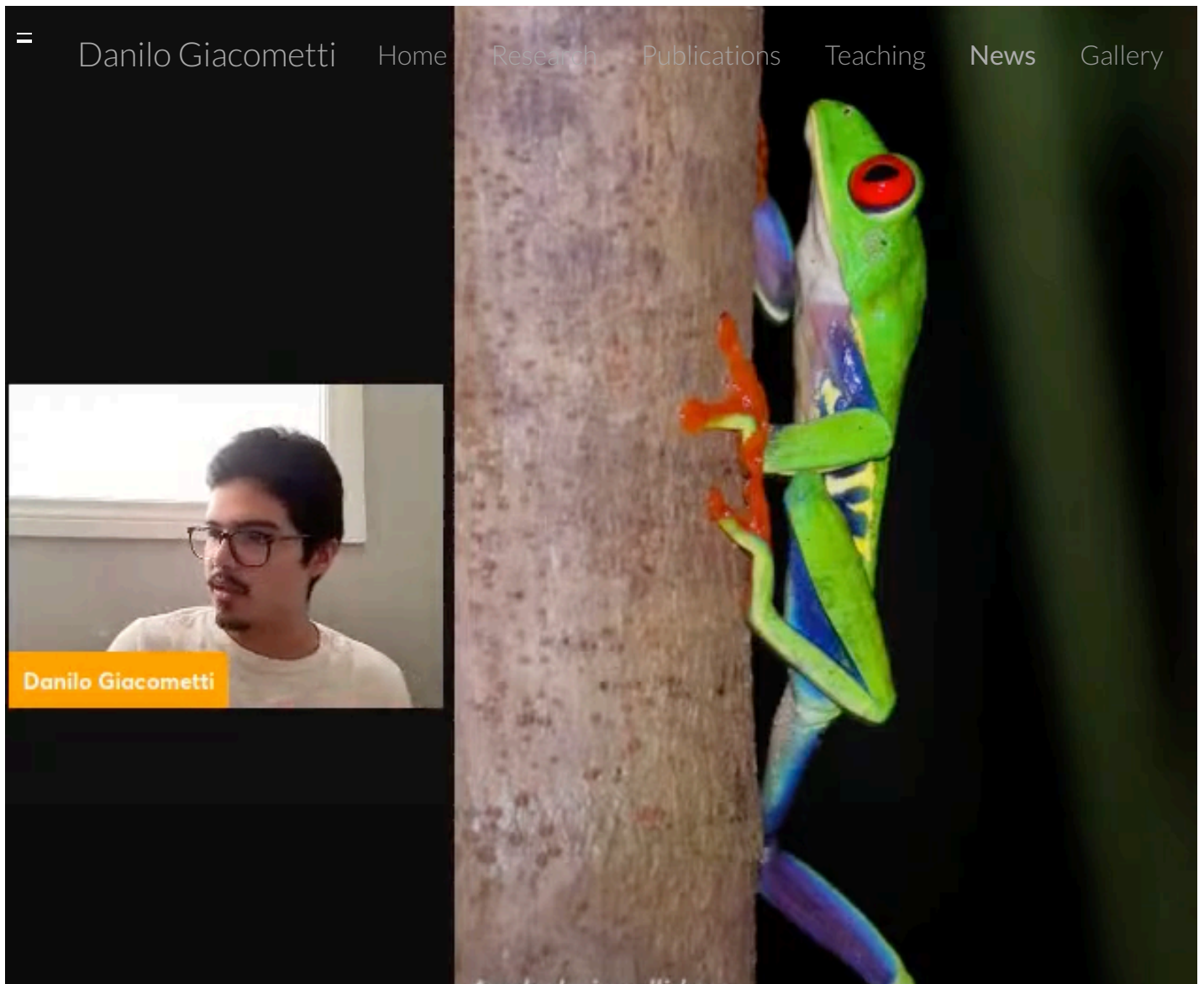
Despite the evidence against thermotaxis in fossorial ectotherms, we found that Spotted Salamanders engaged in active behavioural thermoregulation in both the active and overwintering seasons. These behavioural differences translated into a 9.4°C shift in selected temperatures between seasons. I also blogged about this study [here](#).



May, 2024

The annual research report of the Algonquin Wildlife Research Station is out now!
Read about our work on the thermal biology of the Spotted Salamander [here!](#)





November, 2023

I gave a lecture on the thermal dependence of behaviour and physiology in amphibians and reptiles as part of a workshop organised by LaZoo UEMG. The talk is available on Youtube, although only in Portuguese.

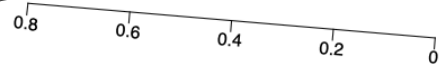


October, 2023

New paper out!

After a long reviewing process, our meta-analysis evaluating the effects of seasonality over lizard thermal biology is finally out in Biological Reviews. I am happy beyond words to see this paper published!





New paper out!

I am really happy to share my first PhD publication. In this paper, we show that fossoriality did not affect the metabolism of amphibians. Instead, body mass, temperature, and shared ancestry explain metabolic differences among species. Make sure to check out my guest blog on the Tattersall Lab website too.



July, 2023

Had a great time at the 2023 JMIH in Norfolk, Virginia, USA. I gave a talk about the relationship between metabolism and fossoriality in amphibians, and we even won trivia night (photo on the left)! Thanks again to the SSAR and the ASIH for funding my attendance to the conference.



May, 2023

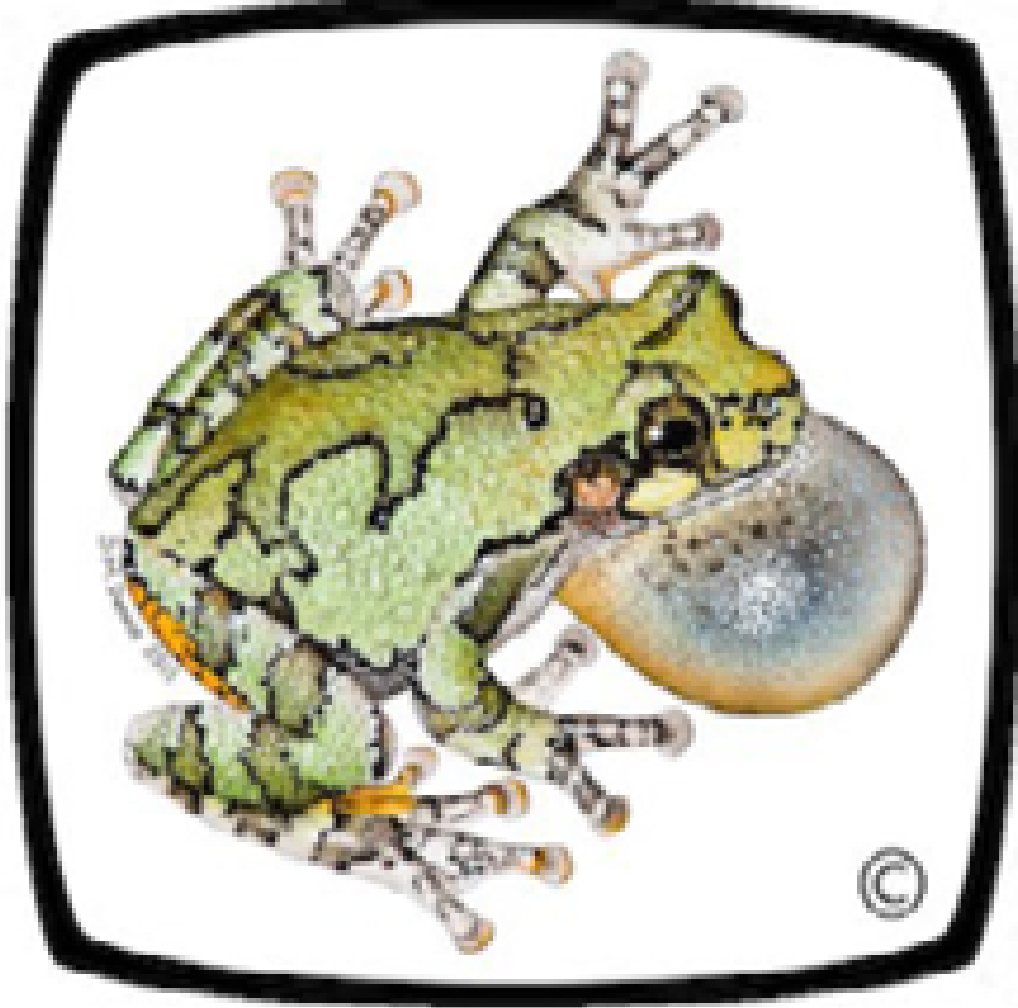
I am happy to share that I am one of the recipients of the Cashner Student Award. Thank you to the American Society of Ichthyologists and Herpetologists for making it possible that I attend the 2023 JMIH in Norfolk, Virginia, USA.



May, 2023

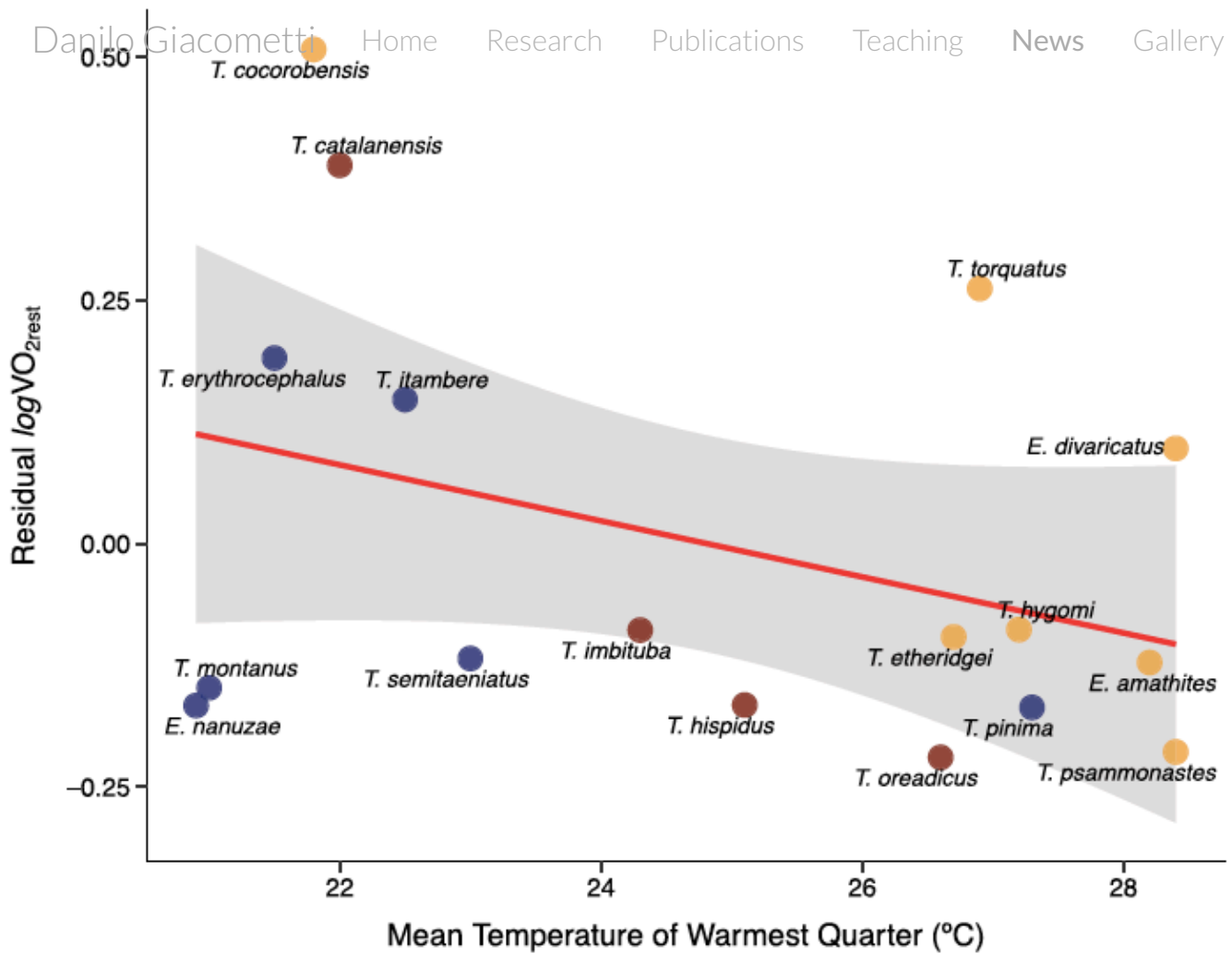
I have just returned from Costa Rica after an amazing week with the Resilience Institute Bridging Biological Training and Research (RIBBiTR) team. There are no words to explain how inspired I feel to keep pursuing my research after meeting so many great people. During this training week, we were able to share our knowledge and ideas in a welcoming environment that gave voice to each and every attendee. When people interested in fostering a better future for science get together, the outcome is always good. Looking forward to keeping in touch with my peers & to learn about their research towards amphibian conservation. Pura vida!





March, 2023

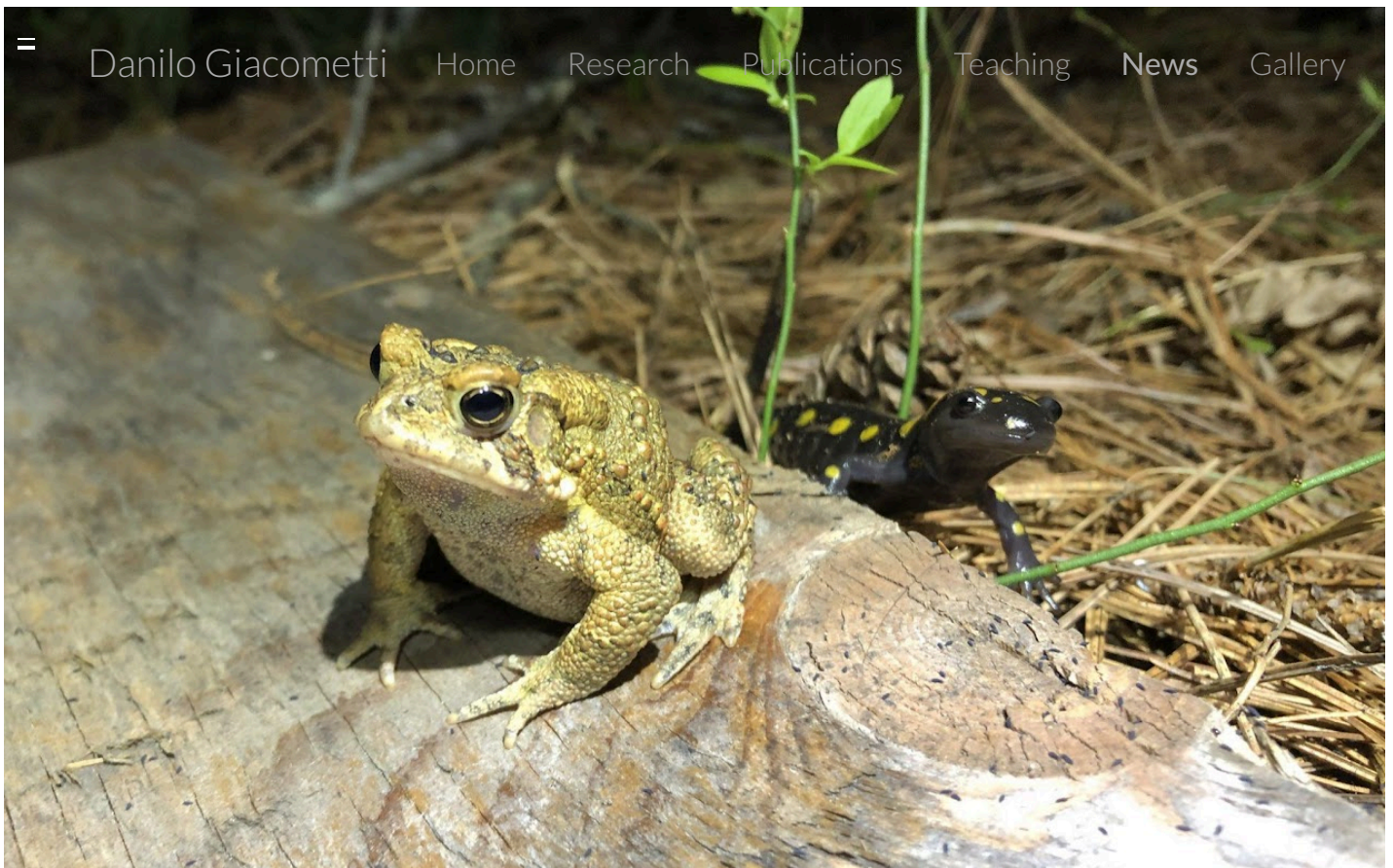
I am happy to share that I am one of the recipients of the Roger Conant grant in the International category. Thank you to the Society for the Study of Amphibians and Reptiles for supporting my work and helping fund my research!



September, 2022

New paper out!

In this study, we found that the interspecific variation in metabolic rates of Tropidurinae lizards is explained by differences in habitat temperature. Species from warmer habitats had lower resting metabolic rates than those from cooler habitats, supporting a mechanism of negative compensation in metabolic responses to temperature.



May, 2022

Field season in Algonquin Provincial Park was fantastic. I had never seen so many amphibians at the same spot in my whole life. Stay tuned for papers on Spotted salamanders coming out soon!

[Danilo Giacometti](#)[Home](#)[Research](#)[Publications](#)[Teaching](#)[News](#)[Gallery](#)

December, 2021

I am happy to share that I am one of the recipients of the Horizon Graduate Scholarship! More information [here](#).

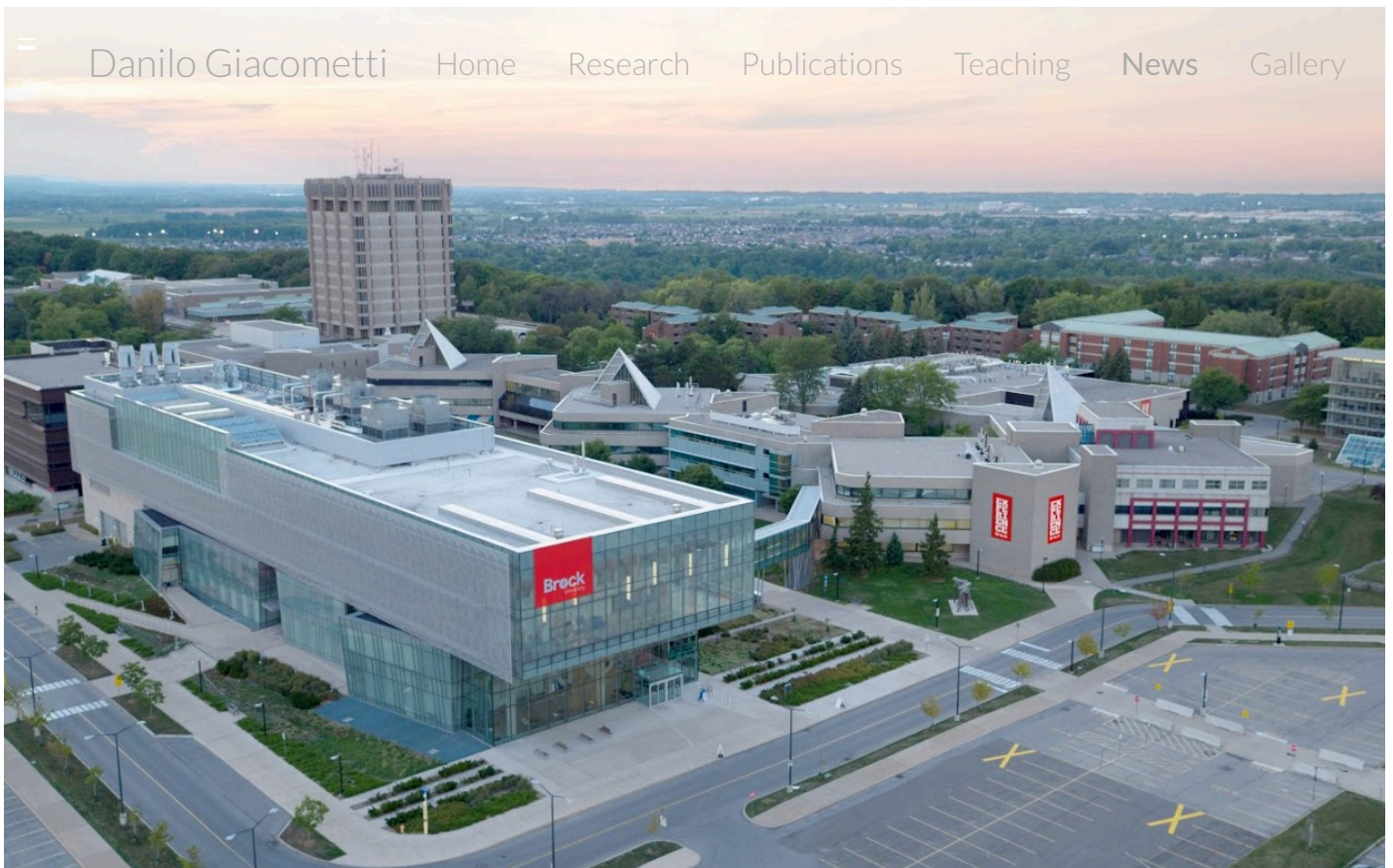




October, 2021

My first paper being part of the Tattersall lab has been published in the Canadian Journal of Zoology!

You can find a link for the pdf [here](#) and a post by Dr. Tattersall [here](#).

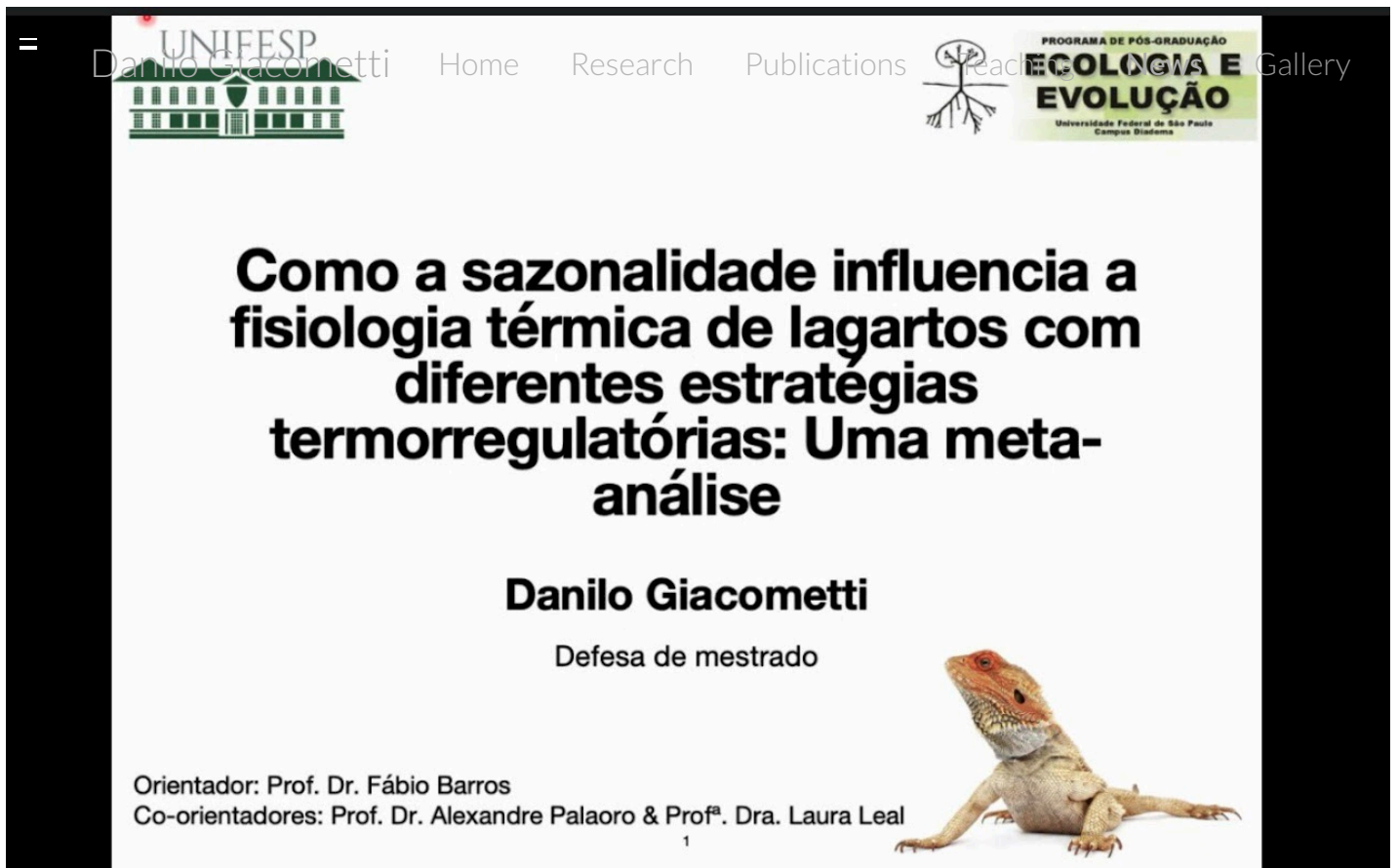


September, 2021

Started my PhD at Brock University supervised by Dr. Glenn Tattersall.

I cannot wait to start taking courses and developing projects in his lab!





September, 2021

I successfully passed my master's defence after over ten days since I arrived in Canada to start my PhD.

Blurred timelines while moving up the academic scale, a dash of anxiety, a virtual defence, but I am glad it all worked out.



December, 2020

First fieldwork in over 11 months. It felt good to be out in the open again. We collected *Tropidurus itambere* at its type locality in Sorocaba, Brasil, which has now unfortunately become a university campus. This was my first time working with a thermal imaging camera, and I cannot wait to use it again!



Como a sazonalidade regula a fisiologia térmica de lagartos?

Danilo Giacometti

Exame de qualificação



Orientador: Prof. Dr. Fábio Barros

Co-orientadores: Prof. Dr. Alexandre Palaoro & Prof^a. Dra. Laura Leal

1

November, 2020

I am now one step closer to defending my master's! I have just presented my preliminary results to the graduate committee (we call this *qualificação* in Brasil), and was told that I am all set to keep conducting my research.

Danilo Giacometti, PhD
Universidade de São Paulo
Departamento de Fisiologia, Instituto de Biociências
Rua do Matão, Travessa 14, 101
Cidade Universitária, 05508-900
São Paulo, Brasil

d.giacometti96@gmail.com